Part 18

Q1. What is Local AI?

Local AI means using an AI model directly on my computer. after downloading the model, I can ask questions and get answers without sending everything to an online AI service.

Q2. What is the difference between a Cloud LLM and a Local LLM?

The main difference is where the model runs. A cloud model runs on a company's servers, while a local model runs on my own machine. With a local model, I can also work offline once everything is installed.

Q3. What role does Ollama play?

Ollama made it easier for me to download and run models like Qwen and Llama on my computer. I also used it to connect my Python code to the models and receive their answers.

Q4. What is model quantization?

Quantization is a way of making a model lighter by storing its numbers with less precision. It helps reduce memory usage, although sometimes it can slightly affect answer quality.

Q5. Why does model size affect memory consumption?

A larger model usually has more parameters to keep in memory. This means it generally needs more RAM or VRAM to run. The precision of the model and the length of the conversation also affect memory usage.

Q6. Why can a smaller model sometimes be preferable?

A smaller model can be easier to run, especially on a normal laptop. It may use less memory and answer faster. In my own tests, Llama 3.2 3B was faster than Qwen3 4B, even though it was the smaller model.

Q7. What are the advantages of local inference?

I like that I can use the model offline and have more control over my data. I can test different models without paying for each request to a cloud API. Of course, privacy also depends on how I configure my application.

Q8. What are the disadvantages of local inference?

The biggest difficulty I noticed was performance. Some responses took several minutes, and larger models needed more computer resources. I also had to install the tools, manage the models.

Q9. Explain the architecture of your Python application.

My project has a Flask web interface, Python code, local course documents, and Ollama. When I type a question, Flask receives it. The application searches the documents for useful information, adds it to the prompt, and sends the request to Ollama. Then it shows the answer, sources, and response time. It also supports conversation history and structured answers.

Q10. Explain the role of the REST API.

The REST API is the connection between my Python application and Ollama. My code sends a question in an HTTP request, and Ollama sends back the model's answer and other information in JSON format.

Q11. What is RAG?

RAG means Retrieval-Augmented Generation. Instead of asking the model to answer only from what it already knows, I first search my local documents for relevant text. I then give that text to the model so it can answer based on the course material.

Q12. What is the difference between an LLM and a multimodal LLM?

A text LLM mainly understands and generates text. A multimodal model can also work with other information, such as images. In this lab, I tested LLaVA by asking it questions about a picture.

Q13. Compare two local models experimentally.

I compared Qwen3 4B and Llama 3.2 3B on my computer. I tested their answers and code in Part 6, then checked their speed and memory use in Part 16. These are my actual results.

Part 6 — Response and code tests

Metric	Qwen3 4B	Llama 3.2 3B
Response time (s)	360.9	32.0
Response quality (/5)	5	4
Code correctness (/5)	5	5

Part 16 — Performance benchmark

Metric	Qwen3 4B	Llama 3.2 3B
Response time (s)	220.93	80.75
Generation speed (tokens/s)	6.70	10.71
Loaded memory (GB)	3.18	2.56
VRAM usage (GB)	0	0

What stood out to me is that Llama was much faster in both tests and used less memory. Qwen received a slightly higher response-quality score in Part 6, but both models got 5/5 for code correctness. So, for my computer, I would prefer Llama for everyday use. I also noticed that response times changed between tests, so I should not judge a model from only one run.

Q14. Which model would you deploy on a computer with limited RAM? Explain.

I would choose Llama 3.2 3B. In Part 16, it used 2.56 GB of loaded memory, compared with 3.18 GB for Qwen3 4B. It was also faster in both experiments, and it still produced correct code in Part 6. For me, it gives a better balance between speed, quality, and memory use.

Q15. What security risks exist when exposing a local LLM server to a network?

If I make my Ollama server available to other computers, people might be able to send requests without my permission because Ollama does not have authentication by default. They could use my computer's resources or potentially expose information from the application. Documents used by RAG can also contain malicious instructions, called prompt injection. I would keep the server on localhost unless remote access is really needed. If I needed to share it, I would protect it with a firewall and an authenticated HTTPS reverse proxy.